Session 4: Interplanetary Transfers and Gravity Assists

Task 1: Explain the patched-conic method and why the excess speed v-infinity is the same going into and coming out of a gravity assist

Patched-Conic Approximation:

- Simplifies a complex multi-body interplanetary trajectory into a series of linked, two-body Keplerian problems by dividing space into distinct **Spheres of Influence (SOI)**.
- Within a planet's SOI $\left(r_{SOI} \approx a\left(\frac{m}{M_\odot}\right)^{2/5}\right)$, gravitational effects of other bodies are neglected, and the trajectory is modeled as a planetocentric hyperbola.
- Outside the SOI, the trajectory transitions seamlessly into a heliocentric Keplerian ellipse centered on the Sun.

Conservation of Hyperbolic Excess Speed (v_∞):

- During an unpowered gravity assist, the spacecraft follows a planet-centered hyperbolic trajectory governed by twobody energy conservation: $\frac{v^2}{2} - \frac{\mu}{r} = \frac{v_\infty^2}{2}$.
- As $r \to \infty$ (at the SOI boundary), the magnitude of excess velocity before entry $(v_{\infty,in})$ and after exit $(v_{\infty,out})$ remains strictly identical:

$$|\boldsymbol{v}_{\infty,in}| = |\boldsymbol{v}_{\infty,out}| = \boldsymbol{v}_\infty$$

- Although v_∞ magnitude is conserved in the planet's reference frame, the gravitational flyby deflects the velocity vector by a turning angle δ, enabling a net heliocentric velocity change $(\Delta \boldsymbol{v}_{helio})$.

Task 2: Inner Transfer (Earth to Venus at $0.723\ AU$)

Using Sun gravitational parameter $\mu_\odot = 1.3271 \times 10^{11}\ km^3/s^2$ and $1\ AU = 1.4960 \times 10^8\ km$

- **Orbital Velocities:**
  - Earth $(1\ AU)$: $v_\oplus = \sqrt{\frac{\mu_\odot}{r_1}} = \sqrt{\frac{1.3271\times10^{11}}{1\times1.4960\times10^8}} = 29.78\ km/s$

- Venus ($0.723\ AU$): $v_{venus} = \sqrt{\frac{\mu_\odot}{r_2}} = \sqrt{\frac{1.3271\times10^{11}}{0.723\times1.4960\times10^{8}}} = 35.03\ km/s$

- **Transfer Ellipse** $\left(a_{trans} = \frac{1+0.723}{2} = 0.8615\ AU\right)$**:**
  - Departure speed at Earth (aphelion):

$$v_{trans,dep} = \sqrt{\mu_\odot\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)}$$

$$= \sqrt{1.3271\times10^{11}\times\left(\frac{2}{1\times1.4960\times10^{8}} - \frac{1}{0.8615\times1.4960\times10^{8}}\right)}$$

$$= 27.29\ km/s$$

  - Arrival speed at Venus (perihelion):

$$v_{trans,arr} = \sqrt{\mu_\odot\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)}$$

$$= \sqrt{1.3271\times10^{11}\times\left(\frac{2}{0.723\times1.4960\times10^{8}} - \frac{1}{0.8615\times1.4960\times10^{8}}\right)}$$

$$= 37.74\ km/s$$

- **Hyperbolic Excess Velocities & Time:**
  - Retrograde departure impulse relative to Earth:

$$v_{\infty,dep} = \left|v_{trans,dep} - v_\oplus\right| = |27.29 - 29.78| = 2.49\ km/s$$

  - Arrival excess speed at Venus:

$$v_{\infty,arr} = \left|v_{trans,arr} - v_{venus}\right| = |37.74 - 35.03| = 2.71\ km/s$$

  - Transfer Duration:

$$t_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu_\odot}} = \pi\sqrt{\frac{(0.8615\times1.4960\times10^{8})^3}{1.3271\times10^{11}}} = 12.62\times10^{6}\ s$$

Task 3: Full Earth-to-Mars Budget

Using $\mu_\oplus = 398600.44\ km^3/s^2$, $\mu_{Mars} = 42828.38\ km^3/s^2$, $R_\oplus = 6378.14\ km$, $R_{Mars} = 3389.5\ km$, and $r_{Mars} = 1.524\ AU$.

- **Heliocentric Transfer Excess Speeds:**

$$v_{\oplus} = \sqrt{\frac{\mu_{\odot}}{r_1}} = \sqrt{\frac{1.3271 \times 10^{11}}{1 \times 1.4960 \times 10^{8}}} = 29.78\ km/s$$

$$v_{Mars} = \sqrt{\frac{\mu_{\odot}}{r_2}} = \sqrt{\frac{1.3271 \times 10^{11}}{1.524 \times 1.4960 \times 10^{8}}} = 24.13\ km/s$$

$$v_{trans,dep} = \sqrt{\mu_{\odot}\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)}$$

$$= \sqrt{1.3271 \times 10^{11} \times \left(\frac{2}{1 \times 1.4960 \times 10^{8}} - \frac{1}{1.262 \times 1.4960 \times 10^{8}}\right)}$$

$$= 32.73\ km/s$$

$$v_{trans,arr} = \sqrt{\mu_{\odot}\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)}$$

$$= \sqrt{1.3271 \times 10^{11} \times \left(\frac{2}{1.524 \times 1.4960 \times 10^{8}} - \frac{1}{1.262 \times 1.4960 \times 10^{8}}\right)}$$

$$= 21.48\ km/s$$

  - Departure excess speed at Earth:

$$v_{\infty,dep} = \left|v_{trans,dep} - v_{\oplus}\right| = |32.73 - 29.78| = 2.95\ km/s$$

  - Arrival excess speed at Mars:

$$v_{\infty,arr} = \left|v_{trans,arr} - v_{Mars}\right| = |21.48 - 24.13| = 2.65\ km/s$$

  - Characteristic Energy (C_3):

$$C_3 = v_{\infty,dep}^2 = (2.95)^2 = 8.70\ km^2/s^2$$

- **Trans-Mars Injection (TMI) from LEO** ($r_{LEO} = 6778.14\ km$)**:**
  - LEO circular speed:

$$v_{LEO} = \sqrt{\frac{\mu_\oplus}{r_{LEO}}} = \sqrt{\frac{398600.44}{6778.14}} = 7.67\ km/s$$

- Hyperbolic injection speed:

$$v_{TMI} = \sqrt{v_{\infty,dep}^2 + \frac{2\mu_\oplus}{r_{LEO}}} = \sqrt{2.95^2 + \frac{2 \times 398600.44}{3778.14}} = 11.24\ km/s$$

- $\Delta v_{TMI} = v_{TMI} - v_{LEO} = 11.24 - 7.67 = 3.57\ km/s$

- **Mars Orbit Insertion (MOI) into** $500 \times 20000\ km$ **Capture Orbit:**
  - Periapsis & Apoapsis radii:

$$r_p = R_{Mars} + 500 = 3389.5 + 500 = 3889.5\ km$$

$$r_a = R_{Mars} + 20000 = 3389.5 + 20000 = 23389.5\ km$$

$$a_{cap} = \frac{r_p + r_a}{2} = \frac{3889.5 + 23389.5}{2} = 13639.5\ km$$

  - Arrival hyperbolic periapsis speed:

$$v_{hyp,peri} = \sqrt{v_{\infty,arr}^2 + \frac{2\mu_{Mars}}{r_p}} = \sqrt{(2.65)^2 + \frac{2 \times 42828.38}{3889.5}} = 5.39\ km/s$$

  - Target capture orbit periapsis speed:

$$v_{cap,peri} = \sqrt{\mu_{Mars}\left(\frac{2}{r_p} - \frac{1}{a_{cap}}\right)} = \sqrt{42828.38 \times \left(\frac{2}{3889.5} - \frac{1}{13639.5}\right)} = 4.35\ km/s$$

  - $\Delta v_{MOI} = v_{hyp,peri} - v_{cap,peri} = 5.39 - 4.35 = 1.04\ km/s$

- **Total Mission Propulsive Budget:**

$$\Delta v_{total} = \Delta v_{TMI} + \Delta v_{MOI} = 3.57 + 1.04 = 4.61\ km/s$$

Task 4: Gravity Assist

Using $\mu_J = 1.266865 \times 10^8\ km^3/s^2$ and $R_J = 71492\ km$ $\left(r_p = 285968\ km\right)$

- **Hyperbolic Eccentricity** (e)**:**

$$e = 1 + \frac{r_p v_\infty^2}{\mu_J} = 1 + \frac{285968 \times 8^2}{1.266865 \times 10^8} = 1.144$$

- **Turning Angle (δ):**

$$\delta = 2 \arcsin\left(\frac{1}{e}\right) = 2 \arcsin\left(\frac{1}{1.144}\right) = 121.9^o$$

- **Maximum Heliocentric Velocity Change ($\Delta v_{helio,max}$):**

$$\Delta v_{helio,max} = 2 v_\infty \sin\left(\frac{\delta}{2}\right) = 2 \times 8 \times \sin\left(\frac{121.9}{2}\right) = 13.99\ km/s$$

Task 5: STK



